

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool: Debugging & Optimisation Task (Low)

Assessment Tool Weightage: 10%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.	BL2 – Understand
2	Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:	BL2 – Understand
3	Correct a faulty SELECT query where column names are misspelled or improperly referenced.	BL1 – Remember
4	Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.	BL3 – Apply
5	Identify and correct a subquery error where multiple values are returned instead of a single value.	BL2 – Understand
6	Debug a VIEW creation statement that contains invalid column references or syntax errors.	BL1 – Remember
7	Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.	BL3 – Apply
8	Identify violations of integrity constraints in a given table and correct them.	BL2 – Understand

9	Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.	BL3 – Apply
10	Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.	BL3 – Apply

Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Judy Allen J

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated

Code Quality	Produces clean, well-structured code with proper comments and	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation
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1. Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.

Original (Faulty):

```
mysql> CREATE TABLE students
->     student_id INT,
->     name VARCHAR,
->     age INT,
->     grade,
->     PRIMARY KEY student_id
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'student_id INT,
name VARCHAR,
age INT,
grade,
PRIMARY KEY studen' at line 2
mysql> |
```

Corrected:

```
mysql> Create table students(
-> student_id INT PRIMARY KEY,
-> name VARCHAR(100) NOT NULL,
-> age INT CHECK (age>=5 AND age<=25),
-> grade CHAR(2)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO students VALUES (1, 'Alice Johnson', 20, 'A');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO students VALUES (2, 'Bob Smith', 22, 'B+');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO students VALUES (3, 'Charlie Brown', 19, 'A-');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO students VALUES (4, 'Diana Prince', 21, 'B');
Query OK, 1 row affected (0.01 sec)

mysql> select * from students;
+-----+-----+-----+-----+
| student_id | name       | age | grade |
+-----+-----+-----+-----+
| 1 | Alice Johnson | 20 | A      |
| 2 | Bob Smith    | 22 | B+     |
| 3 | Charlie Brown | 19 | A-     |
| 4 | Diana Prince | 21 | B      |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Errors Fixed:

1. Missing parentheses around column definitions
2. VARCHAR missing size parameter
3. 'grade' column missing data type
4. PRIMARY KEY syntax incorrect (should be after column definition or with parentheses)

2. Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue.

Original (Faulty):

```
mysql> SELECT department, AVG(salary) FROM employees;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of S
ELECT list contains nonaggregated column 'a3.employees.department'; this is
incompatible with sql_mode=only_full_group_by
mysql> |
```

Corrected:

```
mysql> SELECT department, AVG(salary) AS avg_salary
-> FROM employees
-> GROUP BY department;
+-----+-----+
| department | avg_salary |
+-----+-----+
| IT         | 78500.000000 |
| HR         | 57500.000000 |
| Finance    | 68000.000000 |
+-----+-----+
3 rows in set (0.00 sec)
```

Error Fixed: Added GROUP BY department to aggregate salaries by department

3. Correct a faulty SELECT query where column names are misspelled or improperly referenced.

Original (Faulty):

```
mysql> SELECT emp_name, depatment, sallary FROM employees;
ERROR 1054 (42S22): Unknown column 'emp_name' in 'field list'
mysql> |
```

Corrected:

```
mysql> SELECT name, department, salary FROM employees;
+-----+-----+-----+
| name          | department | salary |
+-----+-----+-----+
| John Doe      | IT         | 75000.00 |
| Jane Smith    | HR         | 55000.00 |
| Mike Johnson  | IT         | 82000.00 |
| Sarah Williams | Finance    | 68000.00 |
| David Brown   | HR         | 60000.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

Error Fixed: Corrected column names: emp_name→name, depatment→department, sallary→salary

4. Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.

Original (Faulty):

```
mysql> SELECT students.name, courses.course_name
      -> FROM students
      -> JOIN enrollments;
ERROR 1054 (42S22): Unknown column 'courses.course_name' in 'field list'
mysql> |
```

Corrected:

```
mysql> select studentss.name, courses.course_name
      -> from studentss
      -> Join enrollments On studentss.student_id = enrollments.student_id
      -> Join courses on enrollments.course_id = courses.course_id;
+-----+-----+
| name   | course_name |
+-----+-----+
| Alice  | Math        |
| Alice  | Science     |
| Bob    | English     |
| Charlie| Math        |
+-----+-----+
4 rows in set (0.00 sec)
```

Error Fixed: Added proper JOIN conditions

5. Identify and correct a subquery error where multiple values are returned instead of a single value

Original (Faulty):

```
mysql> SELECT name FROM studentss
      -> WHERE student_id = (SELECT student_id FROM enrollments WHERE course_id = 101);
ERROR 1242 (21000): Subquery returns more than 1 row
mysql> |
```

Corrected:

```
mysql> SELECT name FROM studentss
      -> WHERE student_id IN (SELECT student_id FROM enrollments WHERE course_id = 101);
+-----+
| name   |
+-----+
| Alice  |
| Charlie|
+-----+
2 rows in set (0.00 sec)
```

Error Fixed: Changed '=' to 'IN' when subquery returns multiple values

6. Debug a VIEW creation statement that contains invalid column references or syntax errors.

Original (Faulty):

```
mysql> CREATE VIEW student_courses AS
-> SELECT s.name, c.course_name
-> FROM students s
-> JOIN enrollments e ON s.student_id = e.id
-> JOIN courses c ON e.course_id = c.course_code;
ERROR 1054 (42S22): Unknown column 'c.course_code' in 'on clause'
mysql> |
```

Corrected:

```
mysql> CREATE VIEW student_courses AS
-> SELECT s.name, c.course_name
-> FROM students s
-> JOIN enrollments e ON s.student_id = e.student_id
-> JOIN courses c ON e.course_id = c.course_id;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM student_courses;
+-----+-----+
| name          | course_name |
+-----+-----+
| Alice Johnson | Math        |
| Charlie Brown | Math        |
| Alice Johnson | Science     |
| Bob Smith     | English     |
+-----+-----+
4 rows in set (0.00 sec)
```

Errors Fixed:

1. Corrected e.id to e.student_id
2. Corrected c.course_code to c.course_id

7. Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.

Original (Faulty - Unnecessary Conditions):

```
mysql> SELECT * FROM products
-> WHERE price > 100
-> AND price >= 50
-> AND category = 'Electronics'
-> OR category = 'Electronics';
```

product_id	name	category	price
1	Laptop	Electronics	1200.00
2	Headphones	Electronics	80.00
4	Monitor	Electronics	350.00

3 rows in set (0.00 sec)

Corrected:

```
mysql> SELECT * FROM products
-> WHERE price > 100 AND category = 'Electronics';
```

product_id	name	category	price
1	Laptop	Electronics	1200.00
4	Monitor	Electronics	350.00

2 rows in set (0.00 sec)

Optimisation: Removed redundant conditions (price >= 50 was redundant with price > 100, and duplicate category condition)

8. Identify violations of integrity constraints in a given table and correct them.

Original (Faulty Data):

```
mysql> CREATE TABLE orders (  
->   order_id INT PRIMARY KEY,  
->   customer_id INT NOT NULL,  
->   order_date DATE,  
->   status VARCHAR(20) CHECK (status IN ('Pending', 'Shipped', 'Deliv  
ered'))  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> INSERT INTO orders VALUES (1, NULL, '2024-01-01', 'Processing');  
ERROR 1048 (23000): Column 'customer_id' cannot be null  
mysql> INSERT INTO orders VALUES (1, 101, '2024-01-02', 'Shipped');  
Query OK, 1 row affected (0.01 sec)
```

Corrected:

```
mysql> INSERT INTO orders VALUES (1, NULL, '2024-01-01', 'Processing');  
ERROR 1048 (23000): Column 'customer_id' cannot be null  
mysql> INSERT INTO orders VALUES (1, 101, '2024-01-02', 'Shipped');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO orders VALUES (1, 101, '2024-01-01', 'Pending');  
ERROR 1062 (23000): Duplicate entry '1' for key 'orders.PRIMARY'  
mysql> INSERT INTO orders VALUES (2, 102, '2024-01-02', 'Shipped');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO orders VALUES (3, 103, '2024-01-03', 'Delivered');  
Query OK, 1 row affected (0.01 sec)
```

Errors Fixed:

1. customer_id cannot be NULL (NOT NULL constraint)
2. status must be one of: 'Pending', 'Shipped', 'Delivered' (CHECK constraint)
3. order_id must be unique (PRIMARY KEY constraint)

9. Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.

Original (Violating 1NF - Repeating Groups):

```
mysql> CREATE TABLE employee_projects (  
->     emp_id INT,  
->     emp_name VARCHAR(100),  
->     project1 VARCHAR(50),  
->     project2 VARCHAR(50),  
->     project3 VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

Corrected (1NF & 2NF):

```
mysql> CREATE TABLE employees (  
->     emp_id INT PRIMARY KEY,  
->     emp_name VARCHAR(100)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> CREATE TABLE projects (  
->     project_id INT PRIMARY KEY,  
->     project_name VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE TABLE employee_project_assignments (  
->     emp_id INT,  
->     project_id INT,  
->     assignment_date DATE,  
->     PRIMARY KEY (emp_id, project_id),  
->     FOREIGN KEY (emp_id) REFERENCES employees(emp_id),  
->     FOREIGN KEY (project_id) REFERENCES projects(project_id)  
-> );  
Query OK, 0 rows affected (0.07 sec)  
  
mysql> INSERT INTO employees VALUES (1, 'John Doe'), (2, 'Jane Smith');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO projects VALUES (101, 'Alpha'), (102, 'Beta'), (103, 'Gamma');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO employee_project_assignments VALUES (1, 101, '2024-01-01');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO employee_project_assignments VALUES (1, 102, '2024-01-15');  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO employee_project_assignments VALUES (2, 103, '2024-02-01');  
Query OK, 1 row affected (0.01 sec)
```

Improvements:

- 1NF: Eliminated repeating groups (project1, project2, project3)
- 2NF: Created separate tables to remove partial dependencies

10. Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.

Original (Redundant Data):

```
mysql> CREATE TABLE order_details (  
->     order_id INT,  
->     product_id INT,  
->     product_name VARCHAR(100),  
->     product_price DECIMAL(10,2),  
->     quantity INT,  
->     customer_id INT,  
->     customer_name VARCHAR(100),  
->     customer_city VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

Corrected (3NF Normalization):

```
mysql> CREATE TABLE customers (  
->     customer_id INT PRIMARY KEY,  
->     customer_name VARCHAR(100),  
->     customer_city VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE TABLE products (  
->     product_id INT PRIMARY KEY,  
->     product_name VARCHAR(100),  
->     product_price DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE TABLE orders (  
->     order_id INT PRIMARY KEY,  
->     customer_id INT,  
->     order_date DATE,  
->     FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
-> );  
Query OK, 0 rows affected (0.10 sec)  
  
mysql> CREATE TABLE order_items (  
->     order_id INT,  
->     product_id INT,  
->     quantity INT,  
->     PRIMARY KEY (order_id, product_id),  
->     FOREIGN KEY (order_id) REFERENCES orders(order_id),  
->     FOREIGN KEY (product_id) REFERENCES products(product_id)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> INSERT INTO customers VALUES (1, 'Alice Johnson', 'New York');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO customers VALUES (2, 'Bob Wilson', 'Los Angeles');  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO products VALUES (101, 'Laptop', 1200.00);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO products VALUES (102, 'Mouse', 25.00);  
Query OK, 1 row affected (0.01 sec)  
  
mysql>  
mysql> INSERT INTO orders VALUES (1001, 1, '2024-01-01');  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO orders VALUES (1002, 2, '2024-01-02');  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO order_items VALUES (1001, 101, 1);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO order_items VALUES (1001, 102, 2);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO order_items VALUES (1002, 101, 1);  
Query OK, 1 row affected (0.00 sec)
```

Improvements:

- Eliminated redundant customer and product data
- Separated into logical tables (Customers, Products, Orders, OrderItems)
- Achieved 3NF by removing transitive dependencies
- `product_price` depends on `product_id`, not `order_id`

